

PART IB Experimental Methods Notes

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0 Introduction

There are children playing in the streets who could solve some of my top problems in physics, because they have modes of sensory perception that I lost long ago.

— J. Robert Oppenheimer

Physics is an empirical science, it attempts to infer the laws of nature therefore it must be based entirely on experiments. That makes knowing how to perform experiments and present results crucial for all physicists. The course focuses on how to plan, execute, and assess measurements, and how to present the results and convince others.

Measurements in physics often involves four major parts:

System → Device → Signal handling → Data Recording,

where the middle two parts includes a lot of error and the final part may include mistakes if someone did not understand his or her data. Statistical methods to analyze the results are also important, and a good physicist always keep an open mind to others' ideas.

This live-text notes was based on the 2024 Part IB Experimental Methods lectures [1] delivered by Professor Ulrich Schneider, my (future) Part II Director of Studies. Some sections contain information from an additional textbook [2], which Cambridge students can download for free with Raven access.

1 Data Sampling

Consider the output of some system, say the position of a pendulum - simple harmonic oscillator. The function $x = x(t)$ is of course continuous. If we try to draw it down by hanging a pencil beneath the pendulum and move a long paper below at constant rate, That would only give us analog signal. However, analog signals are difficult to handle and store, that's why we need the **ADC** - Analog to digital conversion. For a digital signal, we only record several (t, x) coordinates to map the "real" signal. Obviously, how to choose the sampling rate is the main problem. Other problems include:

- How often should we sample the underlying function (sampling frequency)?
- For how long is each recording (data point) averaged?
- How accurately should we sample the underlying function (vertical resolution)?
- How long in total should we sample the signal for?

We shall figure them out one by one.

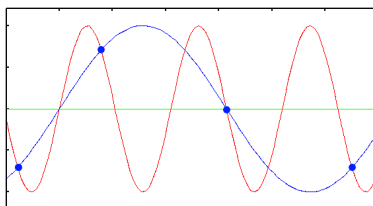


Figure 1: An example of undersampling.

First, let's take a look at sampling frequency. Consider the case shown in Fig. 1, it is obvious that we didn't take enough data points, but how many data points are suitable? To answer this, we need to look at the spectrum of that function, using Fourier Transforms.

1.1 Fourier transforms

Using Fourier transforms

$$\begin{aligned} g(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt; \\ f(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{i\omega t} d\omega, \end{aligned} \tag{1}$$

we could better store data while keeping all of the information from it. For example, for the harmonic oscillator $x(t) = A \cos \omega_0 t$, we have

$$\begin{aligned} g(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} A \cos \omega_0 t e^{-i\omega t} dt \\ &= \frac{1}{\sqrt{2\pi}} \frac{A}{2} \int_{-\infty}^{\infty} (e^{-i\omega_0 t} + e^{i\omega_0 t}) e^{-i\omega t} dt \\ &= \frac{1}{\sqrt{2\pi}} \frac{A}{2} [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]. \end{aligned} \tag{2}$$

The spectrum is therefore two scaled delta functions at $\pm\omega_0$.

Another example is the sinc function, which frequently appears in optics and quantum mechanics,

$$\text{sinc}(t/T) = \frac{\sin(\pi t/T)}{\pi t/T}.$$

The Fourier transform of it is too hard to calculate, we shall do the reverse. Since we've known from diffraction that the image of a slit is a sinc function, the Fourier transform of sinc should be a rect (top-hat) function. If $g(x) = c$ in $x \in (a, b)$,

$$\begin{aligned}\sqrt{2\pi}\tilde{g}(k) &= \int_a^b ce^{-ikx} dx \\ &= \frac{ic}{k} (e^{-ikb} - e^{-ika})\end{aligned}\tag{3}$$

When $a = -b$, we have

$$\sqrt{2\pi}\tilde{g}(k) = \frac{ic}{k} (e^{ika} - e^{-ika}) = \frac{2c \sin k}{k}.\tag{4}$$

Similarly, we can conclude from diffraction that the Fourier transform of discrete delta functions are also district delta functions, with $\tilde{\Delta} = 1/\Delta$.

1.2 Convolution

The convolution of two functions is defined by

$$h(z) = f * g = \int_{-\infty}^{\infty} f(t)g(z-t) dt,\tag{5}$$

with properties (convolution theorem)

$$\begin{aligned}\mathcal{F}[f(t) \cdot g(t)] &\propto \mathcal{F}[f] * \mathcal{F}[g]; \\ \mathcal{F}[f * g] &\propto \mathcal{F}[f] \times \mathcal{F}[g],\end{aligned}\tag{6}$$

convolution turns into a multiplication under Fourier transformation. Convolution predicts how the function will do later when we know the present. It is commonly used as a tool for removing mosaics or building neuro-network in computer science.

The discrete digital data we collect is just the convolution of discrete delta functions and the FT of the original function $x(\nu)$.

$$s(\nu) = x(\nu) * c(\nu),\tag{7}$$

where $c(\nu)$ is an array of δ -fuctions with spacing $\Delta\nu = 1/T$. $S(\nu)$ looks like an array of the spectrum $x(\nu)$, where each spectrum is called a convolution image.

If we want to recover the signal from sampled representation $S(\nu)$, we will need to multiply $S(\nu)$ with a top-hat (box function of width $1/T$). This process only works if the convolution images don't overlap, or we will get **aliasing** images.

The real-time version of this reconstruction is done by replacing multiplication in the Fourier domain with convolution in the time domain (convolving with sinc function by the convolution theorem). That's simply because we cannot get the full spectrum when the experiment is not fully done.

1.3 Nyquist's criterion

Basically, for a band-limited function, one needs to sample at a minimum rate of **twice the highest frequency** component present in the signal. If the sampling is noiseless, one can recover the signal perfectly from its samples in this way.

More in detail, we assume the spectrum of the signal only contains frequency over a known bandwidth, B . If we sample at $\geq 2B$ the convolution images need not overlap, implying that a sampling rate of $\geq 2B$ is acceptable.¹ If there is an overlap due to undersampling, we should either sample faster or lower the signal frequency (shifting the spectrum to lower frequencies).

So, if we want to sample at sub-Nyquist rates, we need to consider what the spectrum of the sampled function will look like to see if overlap of the convolution images will occur. If there is overlap, one should:

¹Note that, B is also in unit Hz, so basically we must sample at a rate quicker than twice the bandwidth.

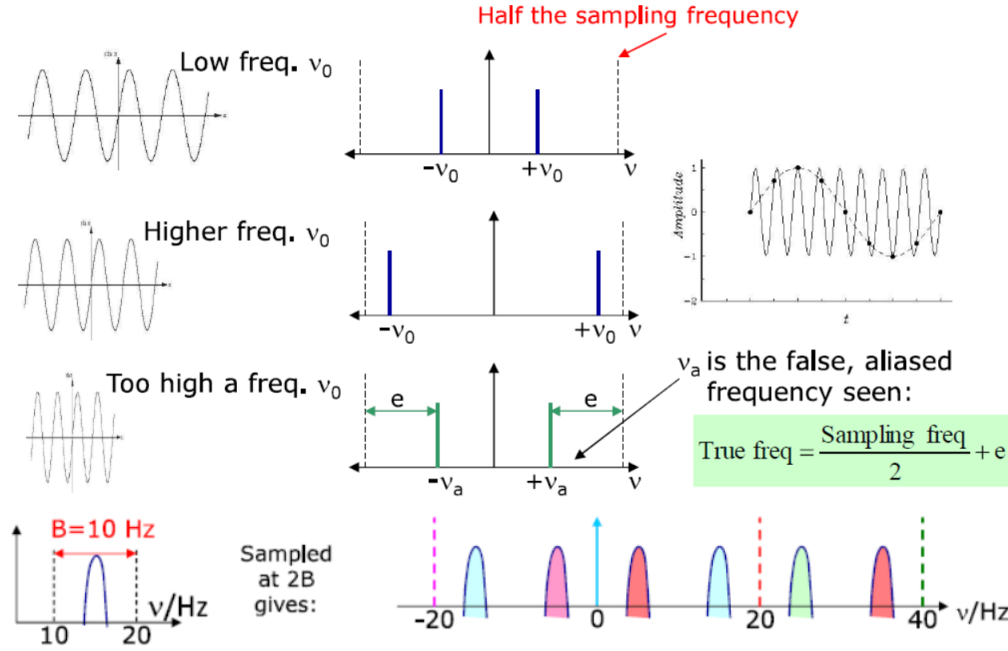


Figure 2: Illustration of aliasing and sampling images.

- Sample faster, or
- Lower the signal frequency, while preserving the information (just shifting the spectrum down). This will not change the bandwidth, but it may be easier to sample faster at these lower signal frequencies.

1.4 Correlations

If we have a continuous signal $f(t)$, the total energy is

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |g(\omega)|^2 d\omega, \quad (8)$$

by Parseval's theorem. We shall consider $|g(\omega)|^2$ as the spectrum energy density, in unit 1/Hz.

The correlation of two signals $f(t)$ and $g(t)$ are defined by the crosscorrelation function:

$$C_{fg}(\tau) = \int_{-\infty}^{\infty} f(t)g(t+\tau) dt. \quad (9)$$

If we want to find out how similar $f(t)$ remains to itself over time, we will need to calculate the autocorrelation function:

$$C_{ff}(\tau) = \int_{-\infty}^{\infty} f(t)f(t+\tau) dt, \quad (10)$$

where τ is the delay time. The autocorrelation function is even in τ , and the value $C_{ff}(0) > 0$ is the maximum value of the function.

It is not so easy to notice that

$$\mathcal{F}[C_{ff}(t)] = |g(\omega)|^2 \quad (11)$$

from the Wiener-Khinchin theorem. It states that autocorrelation function and power spectrum are related by Fourier transform. And the total energy is simply

$$E_{\text{tot}} = C_{ff}(0). \quad (12)$$

Common situation is

$$C_{ff}(\tau) = C_{ff}(0) \exp(-\alpha\tau),$$

where

$$\mathcal{F}[\exp(-\alpha|t|)] = \sqrt{\frac{2}{\pi}} \frac{\alpha}{\alpha^2 + \omega^2}.$$

Spectrum $|g(\omega)|^2$ follows the Lorentzian function.

1.5 Quantization levels

Let's move on and consider how accurately must we sample our function. Digitizing is a two-step process involving sampling and quantizing. The quantization levels are discrete, not continuous. The digitizing process is of course done by the analogue-to-digital converter (ADC). The calculation is related slightly to information theory, but the only thing we need to know is that:

$$\text{Quantizing to } 2^n \text{ levels requires } n \text{ bits.} \quad (13)$$

For example, the ADC in our experiment [3] measures photon electron rates from 0 to about 16 PE/ns, and the data was quantized into $2^{14} = 16384$ levels. Here the quantization error is very low in percentage, and it's even lower in XENONnT because their ADC quantize data to $2^{23} = 8,388,608$ levels, which makes measurements extremely precise.

However, what should we do if we don't have an ADC of that quality (mostly because of insufficient funding)? An answer is oversampling, sampling with higher sampling rate and averaging the results every few points.

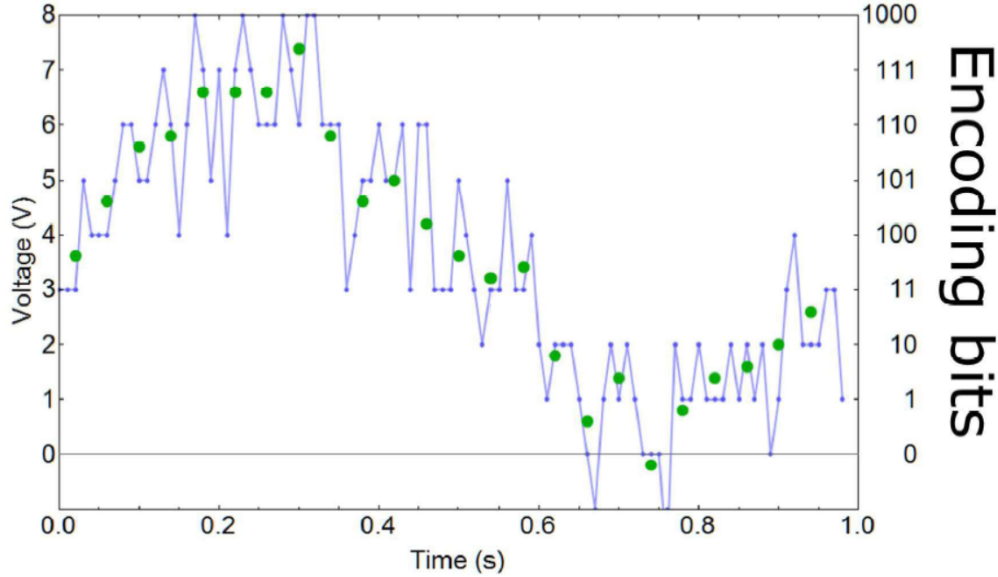


Figure 3: Illustration of oversampling

If the signal contains random noise, then faster sampling means taking more measurements, and so averaging of N multiple samples improves the S/N by a factor of $\sqrt{N}$.

1.6 Proper time range

The third question was, how long should we sample our function for. The answer is of course as long as possible, but with a lower limit of twice per cycle. That means the lower the frequency we are interested in, the longer we have to sample the signal.

A consequence of this is that our spectral resolution will be given by $\Delta f \sim 1/T_{\max}$.

2 Measurements in Physics

Almost every type of measurements of modern physics is turning the quantity we are interested in into currents and voltages. That's why oscilloscopes are so important in physics. The possible sampling frequency of a digital oscilloscope is really wide, normally from 0.01 Hz to 100 MHz. Expensive ones may up to 100 GHz. The voltage levels are also of high resolution.

The only problem is, as an apparatus, it may affect what it is measuring.

2.1 Input and output impedances

Basically, the setup is like this:



Figure 4: Input and output impedances

And by continuity of current,

$$\frac{V_1 - V_{in}}{Z_{out}} = \frac{V_{in} - 0}{Z_{in}} \Rightarrow \frac{V_{in}}{V_{out}} = \frac{Z_{in}}{Z_{in} + Z_{out}}. \quad (14)$$

Normally we use the approximation that $Z_{out}/Z_{in} = 0$, whose reasonableness was verified in the first practical session. Therefore, V_1 is taken as V_{in} .

When the oscilloscope is turned to AC coupling mode, the difference is that it introduces a high-pass filter to the system (R in series with C , and V_{out} is the voltage across R). This is designed to filter out thermal and other kinds of noise at low frequencies. Similarly, a low pass filter can be build by connecting R and C in parallel, and V_{out} is the voltage across them.

To visualise the properties of the filters, we use a kind of plot called the Bode plot, which is two sub-plots of gain, G , and phase difference, ϕ , against log scale frequency.

The high Z_{in} of an oscilloscope (normally higher than 1 M Ω) is great for measuring voltage signals, but little power gets into the oscilloscope. In general, we frequently amplify and/or modify them by using EE units such as operational amplifiers (op-amps).

2.2 Operational Amplifiers

Op-amps look like spiders because they have eight "feet", it is packaged and we (even engineers) do not care what is inside. It is just used as a black box - we know it has some properties, and we use these properties to assist us.

Here is a planar view of an op-amp:

We model it as an ideal voltage amplifier. First we say that,

$$V_{out} = AV_{in}, \quad (15)$$

unless the op-amp saturates ($V_{in} \geq V_{power}$). Assuming $A = \infty$, $Z_{in} = \infty$ and $Z_{out} = 0$, we have our golden rules for analysing op-amps:

- The inputs draw no current, that is, there is no current at the inverting pin '-'.

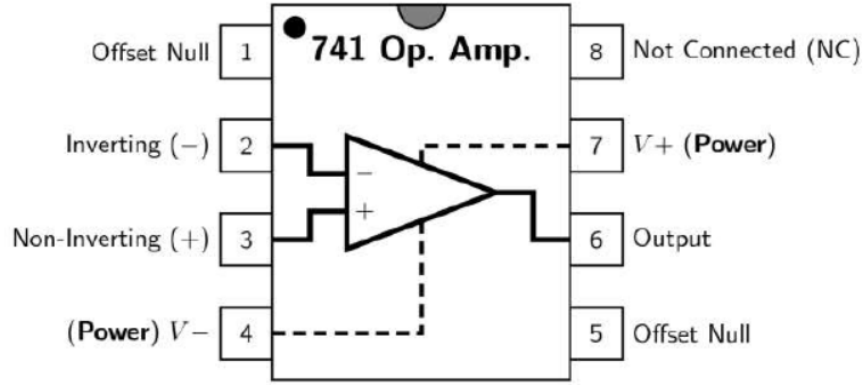


Figure 5: A simplified view of an op-amp, the small black dot at the top left corner indicates its direction.

- The voltages on the input pins are the same, unless the output is saturated².

The inverting amplifier has a gain

$$G = \frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}. \quad (16)$$

It looks like:

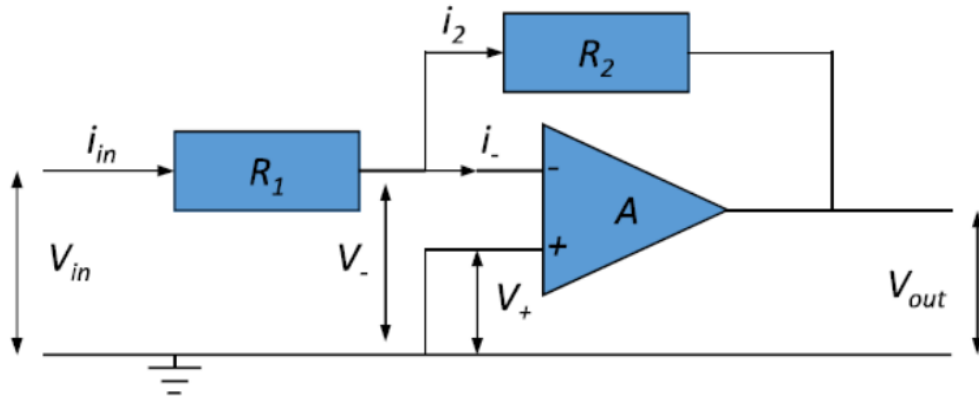


Figure 6: The inverting amplifier, negative feedback is applied by connecting the inverting pin and output using a resistor R_2 .

Similarly, non-inverting amplifier is built by connecting the inputs into the non-inverting pin, and provide a negative feedback through the inverting pin. It has a gain of

$$G = \frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1}, \quad (17)$$

which is also controlled by resistors.

Op-amps can do much more things than amplify voltages. For example, it can be used as a frequency selector if we change the R_2 in Fig. 6 to a $LR \parallel C$ circuit.

²This requires negative feedback.

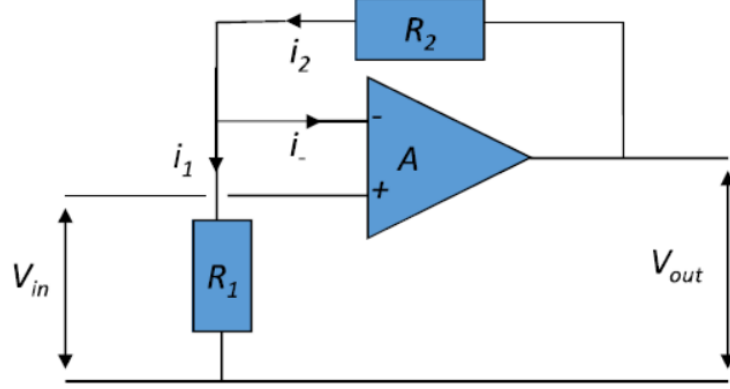


Figure 7: A non-inverting amplifier.

2.3 Non-ideal op-amps

We can justify the golden rules by considering a finite A , consider an inverting op-amp,

$$\begin{aligned}
 V_{\text{out}} &= A(V_+ + V_-) = -AV_-; \\
 i_{\text{in}} &= i_2; \\
 \Rightarrow V_{\text{out}} &= -\frac{R_2}{R_1} V_{\text{in}} \left(\frac{A}{A + \left(1 + \frac{R_2}{R_1}\right)} \right); \\
 V_+ - V_- &\simeq -\frac{1}{A} \frac{R_2}{R_1} V_{\text{in}} \simeq 0.
 \end{aligned}$$

For a real op-amp, A is about 10^4 to 10^6 , Z_{in} is very high, and Z_{out} is very low. Also, A could be complex and frequency-dependent. The amplifying circuitry has a finite slew rate $\partial V_{\text{out}}/\partial t$.

Furthermore, there is an input bias current independent of V_{in} (up to 10^{-7} A) that sets an upper limit to external resistor values. There is also an output voltage independent of $(V_+ - V_-)$ (up to 10^{-2} V) that must be balanced with an external potentiometer.

In the transportation of signal, there are delays which cause phase differences in the Fourier domain,

$$\mathcal{F}[f(t - \tau)] = F(\nu)e^{-i2\pi\nu\tau}, \quad (18)$$

where τ is the time delay.

Examined the frequency dependency in practicals³, we found that op-amps have build-in low-pass behaviour to avoid becoming unstable at high frequencies. This leads to several consequences.

For example, consider switching a circuit on, the step has Fourier components at all frequencies, so positive feedback is bound to happen. So, op-amps are designed to have "internal frequency compensation" so that A itself falls at high frequencies. This is still not always sufficient. We also need to stop oscillation-inducting signals getting back to the inputs, such as the signals picked up from EM radiation, or the inducting signal along supply lines⁴.

3 Control of a System

The basic control of the system is applied in the form of a control loop, different methods would lead to different use and efficiency of the controller, but the logic is simple and identical.

- System is given to us, and we are working hard to control it.

³Treating op-amps as black boxes, as we previously mentioned.

⁴This is why we use decoupling capacitors (typically $10 \sim 100$ nF) between every voltage supply and Earth.

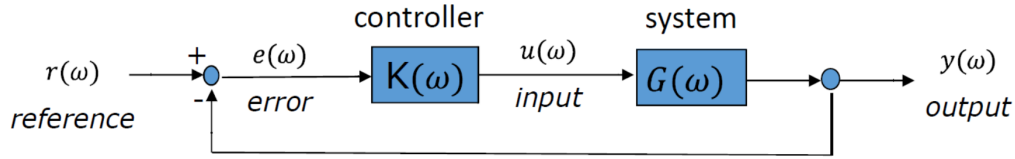


Figure 8: A simple sketch of a control loop.

- We design the controller to control the system.
- Reference r is the desired value.
- Error $e = r - y$ is the difference between desired value and current actual value.
- Input to the system is the output of the controller, $u = Ke$.
- Output of the system is $y = Gu$.

Going around the loop, the loop gain $L = GK$ should be as large as possible. Since,

$$y = \frac{GK}{1 + GK} r. \quad (19)$$

The loop is necessary because it can better deal with noise due to:

- Additional noise added at different stages in the control system;
- Transfer function G might change over time;
- System might be non-linear;
- ...

3.1 Bode plots

A complete Bode plot contains two sub-plots, gain vs frequency and phase difference vs frequency. It is a great way to test how the signal was changed. Here we define several quantities appeared in the diagrams:

- Gain margin: At 180° phase, how far is the gain below 1.
- Phase margin: At the highest frequency with unity gain, how far is phase difference from 180° (inverting).
- Rule of thumb: For a well-behaved system, gain margin should be greater than two and phase margin should be greater than 60° .

3.2 Loop shaping

If we could control the loop gain, how much gain is appropriate? There are many considerations. For example, noise rejection and well tracking need large loop gain, but stability against oscillations needs small loop gain. A typical compromise is that we make the gain high at low frequencies, and low at higher frequencies. This is typically done by adding a low-pass filter.

3.3 Proportional-Integral-Derivative controller

Proportional controller - constant gain K , no phase shift. It is simple and often robust, but very useful.

Integral controller - gain diverges for $\omega \rightarrow 0$,

$$u(t) = K_i \int_0^t e(\tau) d\tau, \quad K_i(\omega) = \frac{K_i}{i\omega}, \quad (20)$$

and has excellent tracking properties $y = r$ for slow changes in r . Often used in slow control systems. But it has a constant phase shift, which reduces phase margin. Also, the low gain at high frequencies means that it is not good at making fast responses.

PI controller - combining P and I,

$$K(\omega) = K_p + \frac{K_i}{i\omega}, \quad u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau. \quad (21)$$

It combines the benefits of P and I, and typically has better performance.

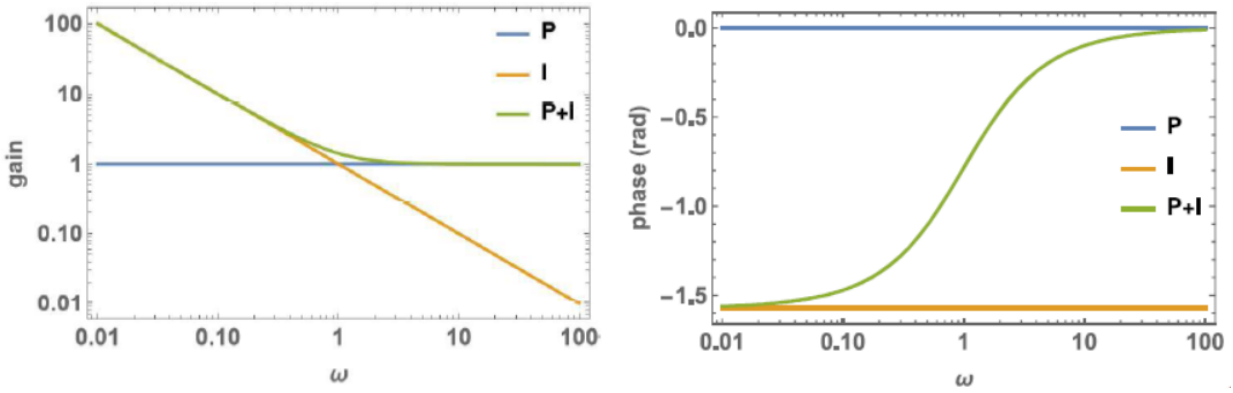


Figure 9: Bode plots of the P, I and PI controllers.

Derivative controller - it reacts to rate of change to error, a good thing is that it can make fast responses, initial value zero is even better in real experiments.

$$u(t) = K_D \frac{\partial e}{\partial t}, \quad K_i(\omega) = K_d i\omega. \quad (22)$$

Its gain diverges for $\omega \rightarrow \infty$, therefore it must be used together with a low-pass filter. Rather comforting, it has positive phase contribution, which can improve the phase margin.

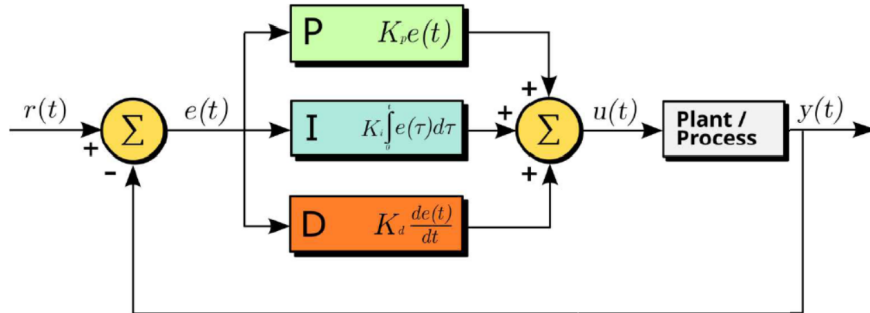


Figure 10: The sketch of PID controller in a control loop.

PID controller - combining everything together, and is the perfect controller that knows the past, present and future. However, tuning the parameters $K_{p,i,d}$ is non-trivial and often involves some trial-and-error, as not all loop parameters are known or keeps constant.

A practical example of PID controller is a self-tuning digital filter applied to a laser system [4].

All previous discussions assumed that the system is linear in the Fourier domain. However, the majority of real systems are non-linear. Our strategies are: If we need to only stabilise around a certain point, we can assume the system is locally linear (Taylor expansion); linearise the system by adding inverse non-linear elements in controller.

4 Errors

In confronting a physical model with an experimental result, we need the experimental value and an estimate of its error to know if the experimental value is significantly different from the theoretical prediction. Therefore error estimation is a critical part of doing physics.

There are two types of errors, random error and systematic error. Random error are treated statistically and it is believed that the average of sufficient measured values is very close to the true value. The source of random error can come from technical or fundamental noise (such as shot noise, which rises from quantisation of nature). Systematic error are roughly constant or drift steadily in time, defined by all errors that are not random. Since it is not random, statistical treatments will not reduce them, only thoughtful designs can do that. This is the part where intelligence really matters.

Random errors control precision, and systematic errors control accuracy.

4.1 How to perform measurements

The number of measurements we should make depends on the desired limit of random error. If the apparatus gives a random error ϵ in a single measurement, measuring N times in total will give a combined error of ϵ/N . Note that, one may not need to measure with the same nominal setup to reveal a mistake, neighbouring data can contribute. Also, the combined error depends on both ϵ and N , therefore upgrading apparatuses or re-design experiments also matters.

4.2 Reducing systematic errors

The best technique is always: use a null method, because almost all detectors are most sensitive around zero value. This was first introduced in the electric bridge.

For high number of measurements on a single value, we can also apply the trick to average every several data points. This can help us to find the time-dependence of some unknown time-dependent errors.

For measurements that have a variable, make sure the measurements of identical setup are done in separate times. This is also used to reduce time-dependent errors.

Measuring the difference in the quantity with the system in the desired state, and then in a standard (if possible, similar states) is also a good idea of reducing systematic errors. Because it is unlikely that many different ways of experiments are wrong by the same value.

For apparatuses such as micrometers, zeroing before measurement means that rigidity is required only on short timescales.

4.3 Selection effects

Here is an example: You have 4 filament lamps. Which radiates the most power? You can measure by look at them and say: "Surely this one is the brightest!" Yes, but a visibly dim one may be still be putting out high power in the infrared region. Selection effects can be subtle and all too easy to miss. You have to make sure you measure what you want to measure. Be extra careful about the results that fits your intuition well, because your thoughts might be wrong.

Understanding, modelling, and minimizing systematic uncertainty is often biggest part of precisions science, an example is the recent measurement of the fine-structure constant [5].

5 Noise

Noise is typically unwanted, unpredictable modifications to a signal. It is assumed to carry no useful information. It typically has a wide frequency spectrum, unless the frequency is fixed (such as the 50 Hz power line or seasonal variances). The type of noise is characterised by the power spectral density

$$\text{psd} \propto |F(\omega)|^2. \quad (23)$$

5.1 White noise

White noise is defined as noise with constant psd. It is fully uncorrelated ($C_{ff}(\tau) = \delta(\tau)$), and a common example is the shot noise on photodiode.

Johnson-Nyquist noise is a kind of white noise in electronic circuits at typical electronic frequencies. It comes from thermal fluctuations proportional to $k_B T$. The solution is easy to think of but hard to do in practice - cool down the electronics in your system. Fine experiments requires to be cooled all the way to $\simeq 0$ K (as cold as possible).

5.2 Brownian noise

Characterised by $\text{psd} \propto \frac{1}{\omega^2}$, Brownian noise comes from temporal integration of white noise or the random walk noise (Gaussian in real space).

5.3 Pink noise

Pink noise is often called $1/f$ -noise because its psd is proportional to $1/\omega$. It is almost ubiquitous, and shows up in many different settings, including most electronics.

Identification of pink noise is done by measuring psd using a spectrum analyser.

5.4 Reducing noise

There are many ways to reduce unwanted noise. The most powerful and expensive one is eliminate the noise at source. For example, the most effective way of removing gravity is sending a satellite into the outer space, but that costs a lot for sure.

Other methods include shielding, differential experiments, and filtering. However, methods such as filtering are removing noise in the control system, which can be ineffective and misleading.

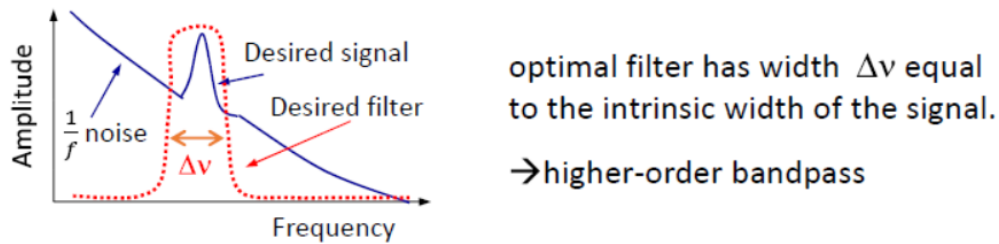


Figure 11: A typical case of filtering.

5.5 Phase-sensitive detection

A easy challenge to find is, what to do if signal is at DC and there is $1/f$ noise? The DC noise tends to infinity and it may completely destroy our signal. The only thing to do is to modulate the signal in time⁵ to bring it to higher frequency and use optimized filter.

⁵Switch on-off periodically.

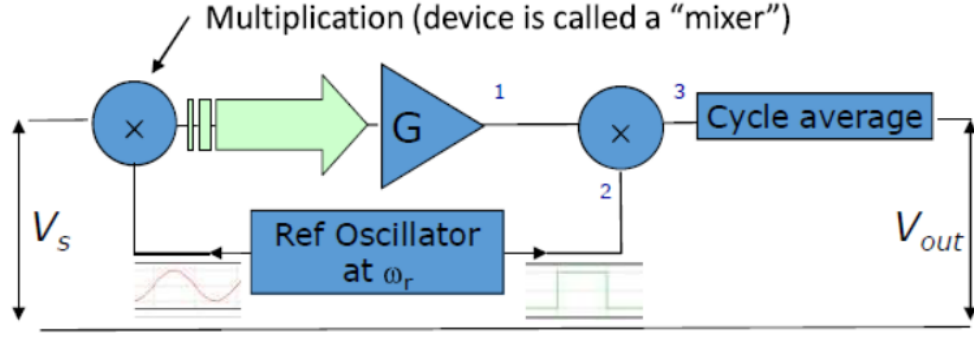


Figure 12: Ref introduces sine wave at 1st modulator. Ref introduces square wave at 2nd modulator. Output of 2nd modulator does not have mean at zero, $V_3 \propto V_S$ and depends on ϕ .

If we changed the square wave in Fig. 12 to an identical sine wave, the voltages are changed to

$$\begin{aligned}
 V_1 &= V_S \sin(\omega_r t + \phi) G \\
 V_3 &= V_1 V_2 = V_S \sin(\omega_r t + \phi) G \sin(\omega_r t) \\
 &= \frac{V_S G}{2} [\cos \phi - \cos(2\omega_r t + \phi)] \\
 \bar{V}_3 &= \frac{1}{2} V_S G \cos \phi.
 \end{aligned} \tag{24}$$

If we add noise into the system before G , only noise very close to ω_r will survive the low-passing. Therefore the system is ideal to suppress $1/f$ noise by using high ω_r .

5.6 Mechanical / vibration noise

Simply reduced by adding springs.

5.7 Thermal noise

Adding floating shield between two surfaces.

5.8 Electrical circuits

Multiple strategies:

- Use bridge to compare resistances.
- Use two strain gauges, one active, the other is used to calibrate for the environment.
- Use twisted pair cable to reduce induction noise.
- Use differential amplifier to ignore all common-mode induced signal.
- Eliminating electrical pickup by Earthing⁶.
- Shield the system by a Faraday cage or μ -metal shield.

⁶Do not create Earth loops!

6 Probability distributions

Probability: degree of belief in the value of quantities/hypotheses. The "Principle of indifference": we assign equal probabilities to events unless we have information distinguishing them. This approach derives from Laplace.

We can assign probabilities on the basis of the relative frequencies of events in the limit of a large number of trials: This is often called the "frequentist" approach. Or, we simply examine what information we have and use that. This is often called the "Bayesian" approach. All observers with the same information must agree on their probabilities.

In physics, there are three commonly-used probability distributions: Gaussian, Binomial, and Poisson distribution.

Binomial distribution is given by

$$P = p^r (1 - p)^{N-r} \frac{N!}{(N-r)!r!}, \quad (25)$$

where p is the probability of "TRUE". The expectation value and standard deviation are

$$\langle r \rangle = Np; \quad \sigma^2(r) = Np(1 - p). \quad (26)$$

For rare events that obeys binomial distribution, $p \rightarrow 0$ and $N \rightarrow \infty$, the probability distribution tends to Poisson distribution,

$$P = \frac{\lambda^r}{r!} e^{-\lambda}, \quad (27)$$

where λ is the event rate (events per unit time interval), and P represents the probability of having r events in the same interval.

$$\langle r \rangle = \sigma^2(r) = \lambda. \quad (28)$$

Since $\sigma/\bar{r} = 1/\sqrt{\lambda}$, shot noise mostly important when dealing with small numbers. When λ is high enough, the probability distribution tends to Gaussian.

$$P = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad \langle x \rangle = \mu, \quad \sigma(x) = \sigma. \quad (29)$$

Sometimes we may see a distribution of Poisson divided by Gaussian, do not confuse it with the Landau distribution [6].

7 Basic Statistics

In this section we want to figure out how to estimate parameters from our data. And how to tell whether our estimation makes sense.

A good example is fitting a straight line, where we have assumed that the x values have no error. If we want to estimate a vector $\vec{a}$ that is related to the slope and intersect of the straight line, we can write the likelihood⁷ as

$$L(y_1, \dots, y_N | \vec{a}) = \prod_i p(y_i | \vec{a}) \, dy_i. \quad (30)$$

One way of estimating $\vec{a}$ is called the maximum likelihood method - find the value of $\vec{a}$ that maximises $L(y_1, \dots, y_N | \vec{a})$. This sounds plausible and is often done, but the likelihood only measures $p(\text{data} | \vec{a})$, and what we really want to maximise is $p(\vec{a} | \text{data})$. The two probabilities are generally different.

Luckily, Bayes' theorem provides a link between the two probabilities.

$$p(\vec{a} | \text{data}) = \frac{p(\vec{a}) p(\text{data} | \vec{a})}{p(\text{data})}. \quad (31)$$

In this equation, $p(\text{data}) = 1$ due to normalisation condition. But $p(\vec{a})$, the *prior* probability of $\vec{a}$ matters. $p(\vec{a})$ incorporates our knowledge of $\vec{a}$ prior to performing the experiment, while $p(\vec{a} | \text{data})$ is called the *posterior* probability for $\vec{a}$, which captures our knowledge of $\vec{a}$ after performing the experiment (obtaining the dataset).

There are three cases of prior knowledge that we need to consider:

⁷Probability of getting our dataset.

- We have no idea what the magnitude of our parameter $\vec{a}$ is. Then a sensible choice is one that is uniform in log space, $p(a) da = 1/a da$.
- We assume that within some range, all values of a are equally likely. In this case the prior probability is a top-hat function. This is the most widely used case.
- If the prior is informative, it should be taken greater care of. A good example is the COVID test with 99.99% chance of giving the true value still can't confirm if a person is infected.

Back to the example of straight-line fitting with flat prior. We shall assume that the errors are Gaussian, with error σ_i . Therefore, for the i -th measurement,

$$p(y_i|\vec{a}) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(y_i - f(x_i|\vec{a}))^2}{2\sigma_i^2} \right], \quad (32)$$

where $f(x_i|\vec{a})$ is the theoretical value obtained from our model (the straight line). The log of likelihood is therefore written as

$$\ln L = -\frac{1}{2} \sum_i \left[\frac{y_i - f(x_i|\vec{a})}{\sigma_i} \right]^2 - \sum_i \ln(\sigma_i \sqrt{2\pi}). \quad (33)$$

We cannot control σ because it is determined by the apparatus and experiment design. What we can do is to maximise the so-called chi-squared

$$\chi^2 = \sum_i \left[\frac{y_i - f(x_i|\vec{a})}{\sigma_i} \right]^2. \quad (34)$$

The only thing left to do is solving the set of equations

$$\frac{\partial \chi^2}{\partial a_i} = 0 \quad \text{for all } i \quad (35)$$

to get the set of parameters.

In other words, the chi-squared is the sum of squares of $y_{\text{actual}} - y_{\text{model}}$, weighted inversely as the error squared. Always remember that it relies on three assumptions:

- The prior is uniform.
- The errors are only in y_i .
- The y_i values are Gaussian distributed about their model values.

For a linear fit, $y = mx + c$, it is easy to compute that

$$\begin{aligned} m &= \frac{\overline{(xy)} - \bar{x} \bar{y}}{\overline{(xx)} - \bar{x} \bar{x}} = \frac{\text{Covariance}(x, y)}{\text{Variance}(x)}; \\ c &= \frac{\overline{(xx)} \bar{y} - \bar{x} \overline{(xy)}}{\overline{(xx)} - \bar{x} \bar{x}} = \bar{y} - m\bar{x}. \end{aligned} \quad (36)$$

The errors in m and c are

$$\begin{aligned} \sigma_m^2 &= \sum_i \left(\frac{\partial m}{\partial y_i} \right)^2 \hat{\sigma}, \quad \hat{\sigma} = \frac{1}{N-2} \sum_i (y_i - (mx_i + c))^2 \\ \sigma_m^2 &= \frac{\hat{\sigma}^2}{N [\overline{(xx)} - \bar{x} \bar{x}]}; \quad \sigma_c^2 = \frac{\hat{\sigma}^2 \overline{(xx)}}{N [\overline{(xx)} - \bar{x} \bar{x}]} \end{aligned} \quad (37)$$

Note that, if the errors are not equal, $\bar{x}$ and $\bar{y}$ must be weighted by $1/\sigma_i^2$.

If our theoretical model f works well, $y - f(x_i)$ should on average equal to σ_i , so χ^2 should approximately be the number of data points N_{data} . If χ^2 is very large, the model is likely to be wrong. If it is very small, one should be suspicious on the current accounts of errors. As the degree of freedom, $N = N_{\text{data}} - N_{\text{parameter}}$, goes to infinity, the chi-squared distribution tends to a normal distribution with variance $2N$.

If the underlying probability distribution is not known, chi-squared test may not be accurate. We need to use non-parametric statistics. Something called "runs test" is a test for detecting non-randomness. Other important non-parametric tests include sign test, Mann-Whitney test, and Kolmogorov-Smirnov test.

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